Sgta 6 2021 - tutorial

Introduction to Mathematics (University of Wollongong)

MATH1015 Calculus and Linear Algebra I (Advanced) Semester 1 2021
SGTA Week 6

1 Class Questions

1. Use elementary row operations to calculate the following determinant:

$$\det \begin{pmatrix} 2 & 4 & 2 & 6 \\ 0 & 0 & 3 & 3 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 0 & 2 \end{pmatrix}$$

   What does your answer tell you about the solution to $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} \in \mathbb{R}^4$?

2. Define two vectors $\mathbf{u} = (4, 2, -3)$ and $\mathbf{v} = (1, -3, -1)$.

   (i) Calculate the angle between these vectors.

   (ii) Show that these vectors satisfy the Cauchy-Schwarz inequality and the triangle inequality.

   (iii) Calculate the unit vectors in the direction of $\mathbf{u}$ and $\mathbf{v}$.

   (iv) Calculate $\text{proj}_{\mathbf{v}}\mathbf{u}$ and $\text{proj}_{\mathbf{u}}\mathbf{v}$.

3. Define $\mathbf{v} = (2, 4)$. Calculate the projection of $\mathbf{v}$ onto the line joining the origin and the point $(-2, -1)$. Draw your result.

4. Define two vectors $\mathbf{u} = (2, 2, 0)$ and $\mathbf{v} = (1, 2, 4)$.

   (a) Find the cross product $\mathbf{w} = \mathbf{u} \times \mathbf{v}$.

   (b) Show that $\mathbf{w}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$.

   (c) Find the area of the triangle with vertices at $(0, 0, 0)$, $(2, 2, 0)$, and $(1, 2, 4)$.

2 Additional problems

Your instructor may discuss some of these problems in the SGTA if time permits, the remaining questions should be completed either before or after the SGTA.

1. Use elementary row operations, or any other appropriate determinant property, to calculate the following:

   (a) $\det\begin{pmatrix}1&4&5&2\\3&0&2&1\\2&2&2&2\\2&2&5&1\end{pmatrix}$ (b) $\det\begin{pmatrix}1&2&3&4\\1&1&3&4\\1&1&1&4\\1&1&1&1\end{pmatrix}$ (c) $\det\begin{pmatrix}5&4&-7&-2\\8&2&-3&-1\\0&1&1&2\\5&4&-7&-2\end{pmatrix}$

   In each case, what can you say about the solution to $A\mathbf{x}=\mathbf{b}$, where $\mathbf{b}\in\mathbb{R}^4$? What can you say about the solution to $A\mathbf{x}=\mathbf{0}$?

2. Define two vectors $\mathbf{u}=(5,2,-3,0,1)$ and $\mathbf{v}=(2,7,1,-3,-1)$.
   (i) Calculate the angle between these vectors.
   (ii) Show that these vectors satisfy the Cauchy-Schwarz inequality and the triangle inequality.
   (iii) Calculate the unit vectors in the direction of $\mathbf{u}$ and $\mathbf{v}$.
   (iv) Calculate $\mathrm{proj}_{\mathbf{v}}\mathbf{u}$ and $\mathrm{proj}_{\mathbf{u}}\mathbf{v}$.

3. Define two vectors $\mathbf{u}=(1,2,3)$ and $\mathbf{v}=(5,2,-2)$.
   (i) Calculate $\mathrm{proj}_{\mathbf{u}}\mathbf{v}$, or the projection of $\mathbf{v}$ onto $\mathbf{u}$.
   (ii) Calculate the unit vector $\hat{\mathbf{u}}$.
   (iii) Calculate $\mathrm{proj}_{\hat{\mathbf{u}}}\mathbf{v}$, or the projection of $\mathbf{v}$ onto $\hat{\mathbf{u}}$. What do you find? Does this make sense?

4. Define two vectors $\mathbf{u}=(4,2)$ and $\mathbf{v}=(1,3)$.
   (i) Calculate the component of vector $\mathbf{v}$ in the direction $\mathbf{u}$ (ie. $\mathrm{proj}_{\mathbf{u}}\mathbf{v}$).
   (ii) Calculate the component of vector $\mathbf{v}$ that is perpendicular to $\mathbf{u}$.
   Hint: Start by finding a vector $\mathbf{w}$ that is perpendicular to $\mathbf{u}$.
   (iii) Draw $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$, $\mathrm{proj}_{\mathbf{u}}\mathbf{v}$ and $\mathrm{proj}_{\mathbf{w}}\mathbf{v}$ on one axis.
   (iv) What do you find if you add $\mathrm{proj}_{\mathbf{u}}\mathbf{v}$ and $\mathrm{proj}_{\mathbf{w}}\mathbf{v}$?

5. Define three vectors in $\mathbb{R}^3$: $\mathbf{u}=(2,-1,2)$, $\mathbf{v}=(4,4,4)$, and $\mathbf{w}=(1,1,0)$.
   (a) Evaluate $\mathbf{u}\times\mathbf{v}$.
   (b) What is the area of the parallelogram that has $\mathbf{u}$ and $\mathbf{u}$ as edges?
   (c) What are the two unit vectors perpendicular to this parallelogram?
   (d) What is the volume of the parallelpiped that has $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ as edges?
   (Bonus: Try to sketch the shape.)

6. Define three points in $\mathbb{R}^3$: $(1,1,1)$, $(2,2,2)$, and $(3,1,3)$.
   (a) What is the vector $\mathbf{u}$ that joins $(1,1,1)$ and $(2,2,2)$?
   (b) What is the vector $\mathbf{v}$ that joins $(1,1,1)$ and $(3,1,3)$?
   (c) What is the area of the parallelogram that has these three points as vertices?
   (d) What are the two unit vectors perpendicular to this parallelogram?

7. Prove that for an $n\times n$ matrix A and scalar k, $\det(kA)=k^n\det(A)$.

8. Define a Cartesian coordinate system as having an x, y, and z plane. Show using a cofactor expansion the cross product of any two vectors in the x-y plane lies on the z-axis. Explain this geometrically.

9. Prove that, given three vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in $\mathbb{R}^3$,

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \cdot (\mathbf{w} \times \mathbf{u}) = \mathbf{w} \cdot (\mathbf{u} \times \mathbf{v}).$$

   Is it also true that $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \mathbf{v} \cdot (\mathbf{u} \times \mathbf{w})$? If not, what does it equal?

   Hint: The scalar triple product is a determinant, so you can apply determinant properties.

10. Prove the Pythagorean identity

$$||\mathbf{u} \times \mathbf{v}||^2 + (\mathbf{u} \cdot \mathbf{v})^2 = ||\mathbf{u}||||\mathbf{v}||.$$

   Hint: There is a cross product property in the notes that involves the angle between the vectors.

11. An $n \times n$ matrix A is created such that each row and each column contains a single one, and ever other term is zero. A 4×4 example is given by

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

   In the previous example, the determinant of A is 1. Show that, no matter where the ones appear, the determinant of A must be either 1 or -1.

   If, instead of ones, each row and column contains one nonzero element. Let p_i denote the nonzero element appearing in row i. Show that the determinant of A must be either

$$\prod_{j=1}^{n} p_j \quad \text{or} \quad -\prod_{j=1}^{n} p_j.$$

   Note: The symbol used here is the "product" symbol, that denotes multiplication of the entries. The notation is similar to the more familiar summation notation, with multiplication used instead of addition. For example, the first product expression represents $p_1 p_2 p_3 \ldots p_n$.